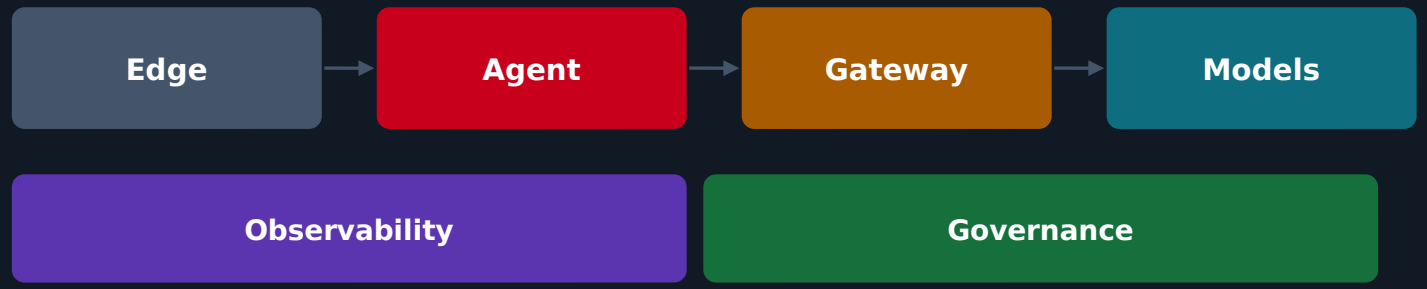
REDCARE PHARMACY · ARCHITECTURE

The Agentic AI Platform, drawn rather than described

Every diagram uses the same six-colour layer key. Learn it once on the next page and the rest of the deck reads itself.



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THE KEY

Six layers, six colours

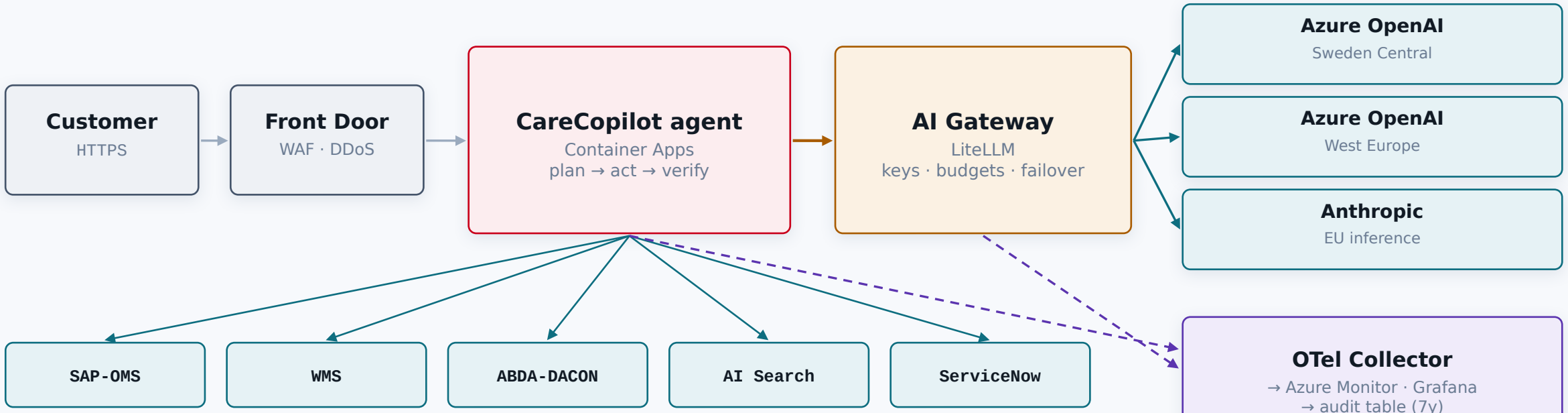
Every box in this deck carries one of these colours. It tells you which layer owns the thing, and therefore who is accountable for it.

Edge `Front Door, WAF`	**Everything hostile arrives here**	Front Door with a WAF terminates TLS, filters OWASP and geo, and absorbs DDoS before anything reaches the VNet.
Agent `Container Apps, the loop`	**Where the reasoning happens**	Container Apps running the plan/act loop. Holds no provider key and cannot reach the internet.
Gateway `LiteLLM`	**The only path to a model**	LiteLLM. Owns virtual keys, entitlement, rate limits, budgets, failover, caching and cost attribution.
Data & AI `OpenAI, Search, Postgres, Redis`	**Models and knowledge**	Azure OpenAI in two EU regions, AI Search for retrieval, Postgres for spend and transcripts, Redis for cache.
Observability `OTel, Monitor, Grafana`	**How the platform sees itself**	OTel to Azure Monitor and Grafana. Four signal families, one trace per turn, an append-only audit table.
Governance `Entra, Key Vault, policy, HITL`	**What keeps it defensible**	Entra ID, Key Vault, policy as code, and the human-approval gate on anything irreversible.

ONE PAGE

The whole system

Solid lines carry a request. Dotted lines carry telemetry. Nothing in the AI plane is reachable from the internet.



systems of record — every tool declares what it touches

Key Vault · Entra ID · managed identity
no long-lived secret exists anywhere in the chain

Human approval on side-effecting tools
the pharmacist ticket does not exist until someone approves it

Edge · Agent · Gateway · Data & AI · Observability · Governance

THE REQUEST PATH

One question, every hop

“Can I take ibuprofen with warfarin?” — and what happens to it, in order.

1 — classify

`costs $0.0001`

A cheap classifier scores the turn trivial, standard or complex.

2 — route

`→ balanced tier`

Entitlement, then tier, then health. The router returns the reason.

3 — budget

`pre-flight`

Tenant daily cap and session ceiling checked before anything is spent.

4 — guard in

`3 checks`

Injection → scope → PII redaction, before the model reads a word.

5 — plan / act

`≤ 6 steps`

LLM proposes a tool, the tool runs, the result comes back. Repeat, bounded.

6 — guard out

`5 checks`

Secrets → grounding → medical-advice policy → PII egress → disclosure.

7 — account

`1 trace`

Spend ledger, four metric families, append-only audit record.

Budget before guardrails

Guardrails cost CPU. A tenant already over budget should be refused before the platform spends anything on them — including its own compute.

Guardrails before the model

Injection has to be caught before the model reads it, and PII removed before the prompt leaves the VNet. An output-only guardrail already lost.

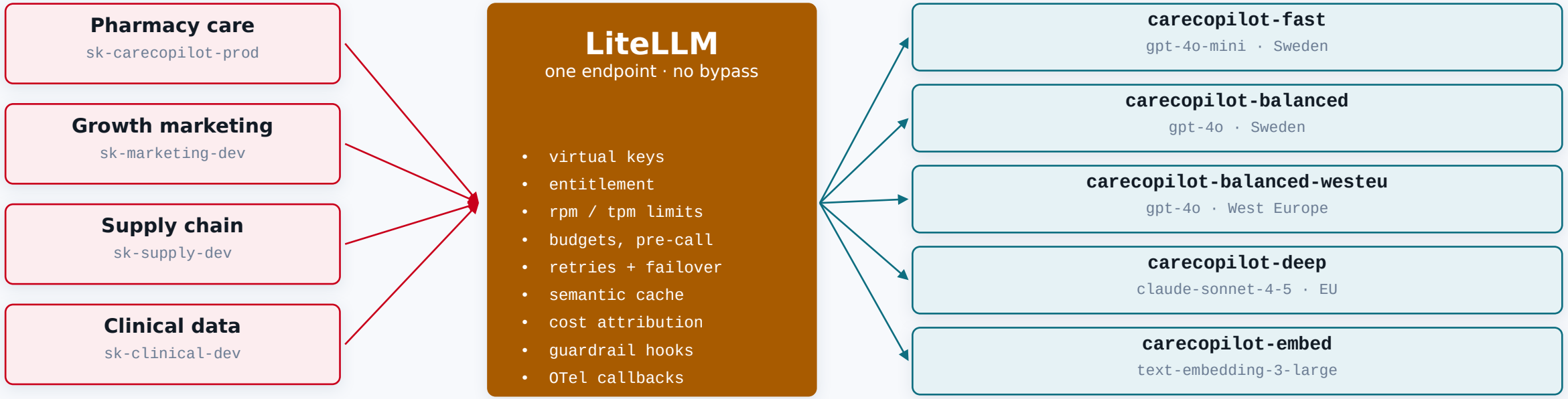
Approval before the side effect

The agent may read and draft freely. It stops before changing a system of record. That is EU AI Act Art. 14 as a code path.

THE CHOKE POINT

Why every call goes through one door

Without it: N codebases holding provider keys, N retry policies, no cost story, no way to revoke a team's access today.



Bypassing it is not a policy — it is a network property. The agent subnet denies internet egress and the agent holds no provider key, so it has exactly one route to a model. Tenants ask for 'carecopilot-balanced', never 'gpt-4o-2024-11-20', which is what makes swapping the model underneath a config change rather than a migration.

ROUTING

Three gates, in this order, always explained

Governance first, cost second, reliability third. Reversing any two of these is a different platform.

1 Entitlement → **2 Complexity tier** → **3 Health**

1 Entitlement

Is this key allowed this model?

A governance question, so it goes first. No performance consideration may override it — that is what makes the entitlement real rather than advisory.

2 Complexity tier

Cheapest model that can do the job

A classifier scores the turn trivial / standard / complex. The mini tier is ~16× cheaper per token; the classifier that decides costs $0.0001 and pays for itself the first time it keeps a greeting off the deep model.

3 Health

Is the chosen deployment cooling down?

Same model in the second EU region first. A different provider is the last resort, because a provider switch changes behaviour and behaviour changes need an eval run.

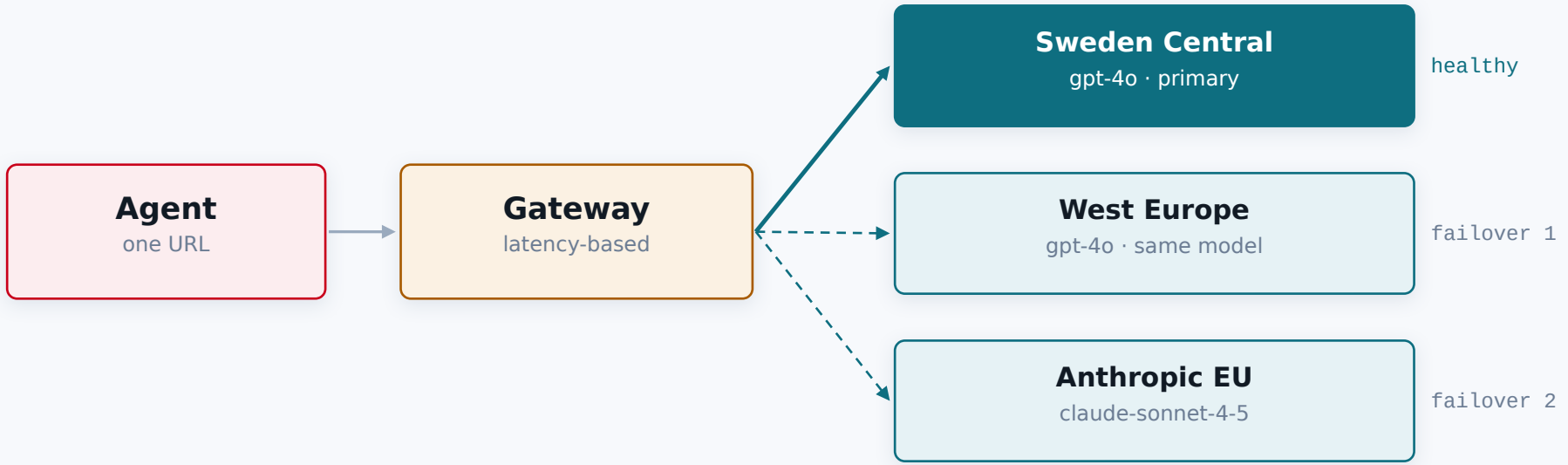
A decision the trace can explain

```
complexity  trivial   —  greeting with no information need
selected    carecopilot-fast   →  azure/gpt-4o-mini @ swedencentral
why         classifier scored the turn 'trivial' → downshift to fast tier
```

RELIABILITY

What happens when a region goes down

Two EU regions and two providers. The second region is failover and a second quota pool — quota is allocated per region.



Region before provider

Same model in a second region keeps behaviour identical. A provider switch does not — so it is last, and it needs an eval run behind it.

3 failures, 60s cooldown

A deployment that fails three times is taken out of rotation for a minute, then probed again. 429s do not count: backpressure is not ill health.

Context and content fallbacks

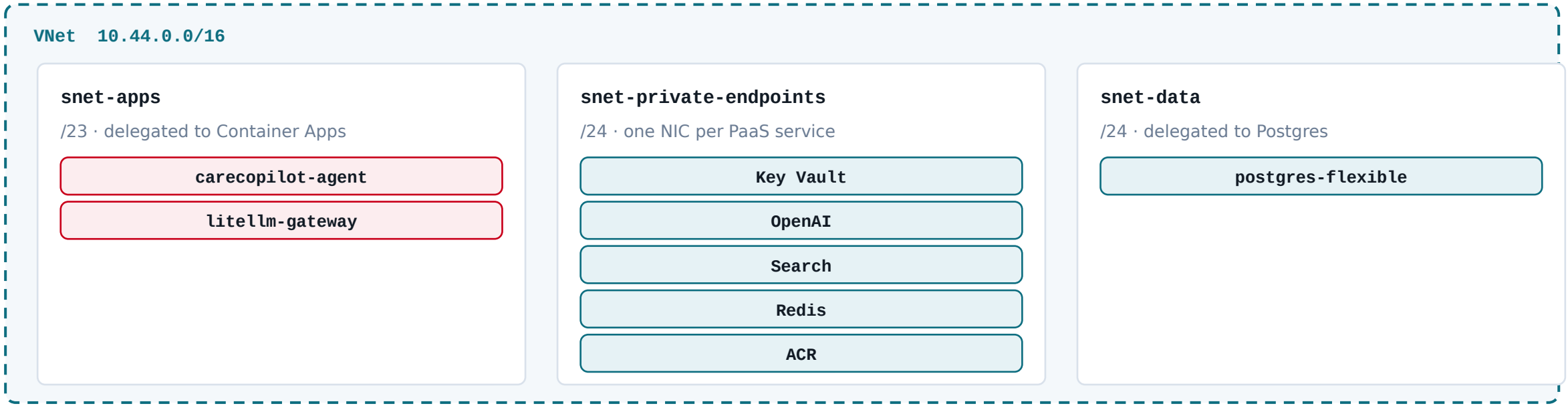
A prompt too long for 128k routes to the 200k model. A provider content-filter block gets a second opinion rather than an error.

A single region is a single point of failure for every AI feature in the company. That is the whole argument for the second one — and it is also where the extra quota comes from.

TRUST BOUNDARY

No public ingress, no unmonitored egress

The rule that shapes every other Azure decision. For a workload handling Art. 9 health data this is the baseline a DPO signs, not hardening.



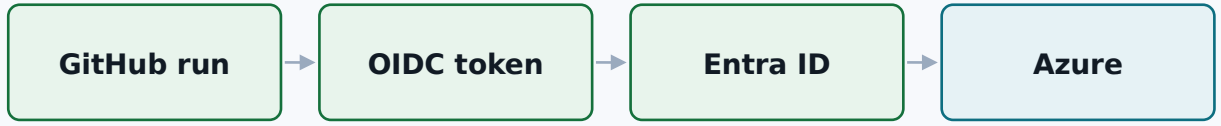
- **Deny inbound from the internet**
  Ingress only from the Front Door subnet, on 443.
- **Deny outbound to the internet**
  Egress allowed to the AzureCloud service tag only — a compromised agent has nowhere to send data.
- **Private endpoint per dependency**
  Key Vault, OpenAI, Search, Redis, Postgres, ACR, Storage, Monitor.
- **A private DNS zone for each one**
  Without it the endpoint resolves to its public IP and traffic silently leaves the VNet — the most commonly missed control in Azure private networking, and it fails quietly.

IDENTITY

There is no long-lived secret anywhere

Two flows, and neither of them ends in a password sitting in a settings page.

GitHub Actions → Azure



Entra checks the token's subject against an exact federated credential — a different branch, environment or fork does not match and is rejected before any Azure call. The token lives minutes. Nothing exists to leak or rotate.

Running app → its secrets

The app holds nothing. It presents a managed identity and Key Vault decides. The gateway and the agent have separate identities, so “the agent was compromised” and “the provider keys were compromised” stay different sentences.

Deploy identity

Contributor on one resource group. Explicitly barred from granting Owner or User Access Administrator — no privilege-escalation path.

Agent identity

Key Vault Secrets User · AcrPull · Search Index Data Reader. Cannot read a provider key. Cannot change infrastructure.

Gateway identity

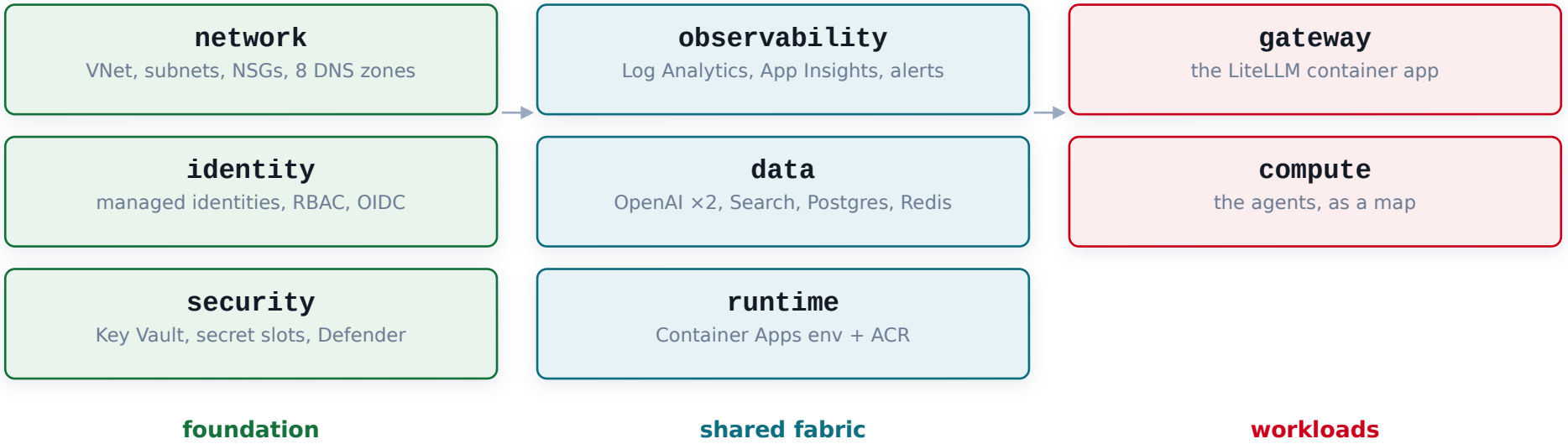
The only identity in the platform permitted to call the model endpoints. That is what makes the gateway's governance non-optional.

The tenant holds a virtual key, not a provider key: revocable, budgeted, scoped, and rotatable without redeploying anything.

INFRASTRUCTURE AS CODE

The module graph is a design signal

Eight modules, two environments, one dependency rule — and the bug that proved the rule.



The cycle that taught the boundary

The gateway needs a registry to pull from; the agents need the gateway's URL. Put the registry in compute and the environment in gateway and the two modules depend on each other — Terraform refuses to build a graph at all. The fix was not a workaround: the compute fabric is shared infrastructure both sit on, so it became its own module. Module boundaries follow ownership, not usage.

dev and prod are the same code

What differs is size and cost: smaller SKUs, no zone redundancy, scale-to-zero. What never differs is posture — private networking, managed identity, no public data access, guardrails on, audit on. If a control had to be switched on for production, dev would be the weak link an attacker uses.

DELIVERY

The only path to production

Nobody holds standing write access to the Azure subscriptions. The pipeline gets minutes of it, via OIDC.

Pull request	Plan on PR	Merge → main	Production
· lint	· terraform plan	· build + SBOM	· canary at 10%
· test	· Checkov + tfsec	· Trivy scan	· bake 15 min vs SLOs
evals ★	· OPA on plan JSON	· cosign sign	· promote or roll back
· terraform	· posted as a comment	· dev deploy + smoke	· 2 approvers
· security			

★ The eval job is a required check

For a non-deterministic system, unit tests prove the plumbing and evals prove the behaviour. Safety is thresholded at 1.00 — one miss fails the build.

The plan is the review artefact

A reviewer approves a diff of resources, not a diff of HCL. Those are different things, and the gap between them is where production surprises live.

Rollback is a traffic weight

Seconds, no rebuild. A rollback that needs a rebuild is not a rollback, it is a hope.

SAFETY

Two pipelines the tenant never writes

Shared, so a team does not have to reimplement PII redaction to ship. They get it by adopting the paved road.

Input pipeline — before the model reads a word

3 checks

`prompt_injection`	`block`	instruction override, exfiltration
`topic_policy`	`block`	outside the pharmacy support scope
`pii_redaction`	`redact`	IBAN, email, phone, card, DOB

Output pipeline — before the customer sees it

5 checks

`secret_leak`	`block`	credential-shaped strings in the answer
`grounding`	`annotate`	every claim traced to a tool observation
`medical_advice`	`escalate`	individualised clinical instruction
`pii_egress`	`redact`	identifiers on the way back out
`disclosure`	`annotate`	EU AI Act Art. 50, on every reply

The trust boundary that matters most in an agentic system

Tool output is data, never instruction. A policy document, an order note or a product description can carry text engineered to look like a system prompt — that is indirect prompt injection, and it is the hard one. There is no complete defence, so the strategy is layered: detect it, treat tool content as data, and constrain the agent so a successful injection has nowhere to go.

OBSERVABILITY

Four families, and one trace with four readers

An agent can be fast, return 200, and still be wrong, ungrounded, unaffordable or looping. Classic APM covers only the first row.

RED

Is the service up?

```
agent_requests_total
agent_request_duration_seconds
agent_errors_total
```

Agent

Is it behaving?

```
agent_steps_per_turn
agent_tool_calls_total
agent_loop_termination_total
```

Quality

Is it right?

```
agent_groundedness_ratio
guardrail_firings_total
agent_eval_score
```

FinOps

Is it affordable?

```
llm_cost_usd_per_turn
llm_cache_events_total
llm_budget_remaining_usd
```

One trace, four readers

Engineering
a debugging artefact

AI team
a quality signal

Finance
a cost line

The DPO
an audit record

PROMISES

SLOs when the output is not deterministic

You cannot promise any individual answer is right. You can promise a rate, measure it, and put an error budget behind it.

OBJECTIVE	TARGET	WINDOW	ERROR BUDGET
Gateway availability	99.9%	30d	43.2 min
Agent turn latency	p95 < 4.0s	30d	43.2 min
Groundedness	≥ 97% grounded	7d	302.4 min
Guardrail coverage	100% of turns	30d	0 min

PAGE **14.4× burn over 1 hour**

2% of the 30-day budget gone within the hour. Somebody wakes up.

TICKET **6× burn over 6 hours**

Degradation that will exhaust the budget this week. A single threshold catches one of these and misses the other.

GOVERNANCE

Four articles, four code paths

A control that exists only in a document drifts. Each of these is a runtime object the service exposes.

Art. 50 **Transparency** `guardrails.ensure_disclaimer`

Every reply carries an AI disclosure and a non-advice notice.

Art. 12 **Record-keeping** `telemetry.record_audit`

Append-only audit table, 7-year retention in the archive tier.

Art. 14 **Human oversight** `orchestrator HITL gate`

Side-effecting tools blocked behind an explicit human approval.

Art. 15 **Accuracy & robustness** `evals.suite.THRESHOLDS`

Groundedness scorer plus a CI eval gate at 0.95.

Risk tier

Limited risk

A customer-facing informational assistant. Not a medical device and not diagnostic, because every clinical judgement is routed to a registered pharmacist — which keeps it out of Annex III.

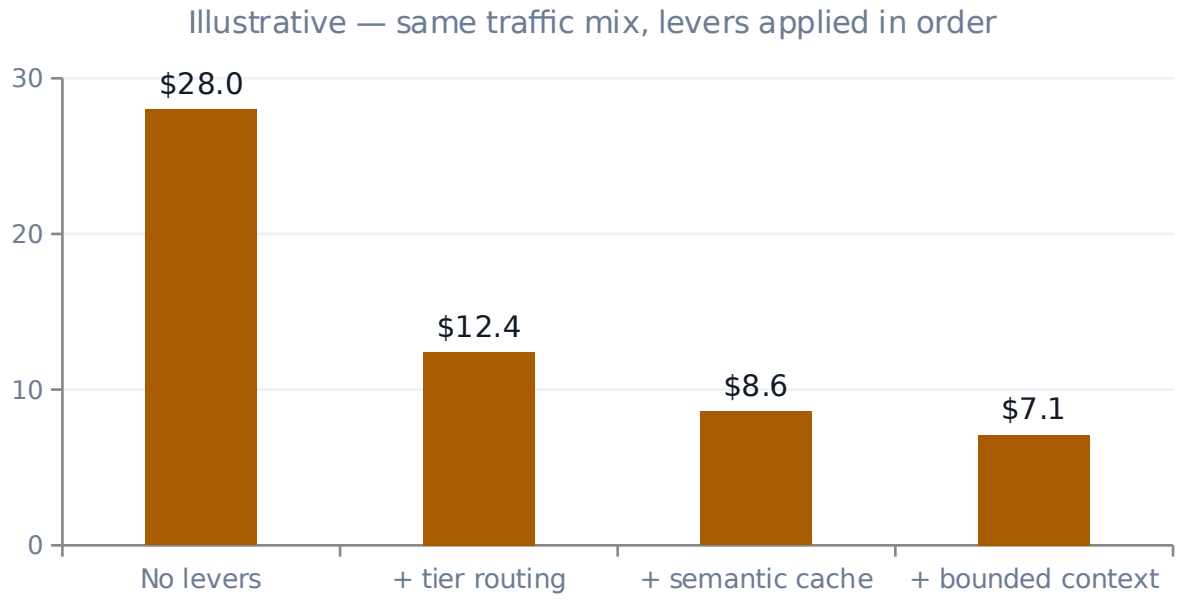
The classification is a decision, made on the record. The moment the agent could give a dose, it is a different tier with a conformity assessment attached.

GDPR: Art. 9 special-category data · basis 6(1)(b) + 9(2)(h) · EU-only residency enforced twice, by a Terraform variable validation and an OPA policy · redaction before the model call · transcripts 90 days, audit 7 years.

UNIT ECONOMICS

Cost per turn is the number

Everything else is a proxy for it. Four levers, in the order I would pull them.



1 Tier routing

A greeting on the mini tier is ~16× cheaper per token than the flagship. The classifier that decides costs $0.0001.

2 Semantic cache

“Where is my order”, “order status?” and “has it shipped” are one question. 25-35% hit rates are ordinary in support.

3 Bounded context

A rolling window stops prompt growth across a long session — the quiet cause of most cost regressions.

4 Hard ceilings

Per request, per session, per tenant per day. A runaway loop stops itself; the gateway is the backstop.

Measured per resolved conversation, not per call — an agent that is cheap per call and never resolves anything is expensive. The reflex to resist in a cost incident is “move everyone to a cheaper model”: that trades a cost problem you can see for a quality problem you cannot.

SCOPE

What is deliberately not here

Each is a “not yet” with a stated trigger, not a “never”. Knowing what you are not building is most of platform product management.

Multi-agent orchestration framework

Two agents that need to coordinate usually mean one workflow was decomposed wrong. Earn it with a real case.

Fine-tuning pipeline

Retrieval and prompting exhaust their headroom long before fine-tuning is the cheapest next move — and it brings a permanent lifecycle with it.

Self-hosted models on GPUs

The economics only work at a volume Redcare does not have yet, and it moves a large operational burden onto a small team.

A bespoke vector database

AI Search covers hybrid retrieval and is already inside the security perimeter.

Kubernetes

Container Apps covers the requirements. AKS is a second product to operate, and the exit stays cheap because the unit of deployment is a plain OCI image.

Agent-to-agent protocols

The standards are still moving. Committing early is expensive to undo.

THE SIXTY-SECOND VERSION

If you remember eight things

The gateway is the whole thesis

It is the only path to a model, which is what makes budgets, entitlement, failover, caching, cost attribution and audit enforceable rather than advisory.

Ordering is the argument

Budget before guardrails. Guardrails before the model. Approval before the side effect.

Gate the side effect, not the reasoning

Read and draft freely; stop before changing a system of record. Art. 14 as a code path.

Evals are the release gate

Tests prove the plumbing, evals prove the behaviour. Safety thresholded at 1.00.

Four signal families, not one

Up, behaving, right, affordable. A platform that ships only RED is trusted until the first bad answer.

Region before provider

Failover keeps behaviour identical first. A provider switch changes behaviour and needs an eval run.

No public ingress, no unmonitored egress

And a private DNS zone for every private endpoint, or the traffic quietly leaves the VNet.

Cost per resolved conversation

Not per call. An agent that is cheap per call and never resolves anything is expensive.